

Pipeline for Optical/infrared Telescopes in Python for Reducing Images (POTPyRI)

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Software

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Summary

Astronomers often need to turn raw telescope images into calibrated, aligned, and combined images before measuring the brightness and positions of stars and galaxies. The Pipeline for Optical/infrared Telescopes in Python for Reducing Images (POTPyRI) automates this process for a number of optical and near-infrared (NIR) telescopes and instruments. It was developed to provide an automated way to reduce imaging data, create a median stack with reliable astrometry and photometric calibration, and perform aperture and Point Spread Function (PSF) photometry on sources within the stacked image.

Statement of need

POTPyRI is written in Python (currently deployed and tested on Python 3.12) and uses packages from Astropy ([Astropy Collaboration et al., 2013, 2018](#)), ccdproc, astroquery ([Ginsburg et al., 2019](#)) and photutils ([Bradley et al., 2024](#)); with the additional use of external dependencies including SExtractor ([Bertin & Arnouts, 1996](#)) and Astrometry.net ([Lang et al., 2010](#)). The code is available on GitHub at <https://github.com/CIERA-Transients/POTPyRI>, where instructions on the installation and detailed use can be found. The code is also installable with pip and is released on PyPI at <https://pypi.org/p/potpyri>. The pipeline has been developed openly on GitHub since 2020, with public releases on PyPI and a public issue tracker for community feedback and contributions.

Currently available instruments (as of March 2026 - see the GitHub for the most recent list) include: - MMIRS (MMT) ([McLeod et al., 2012](#)) - Binospec (MMT) ([Fabricant et al., 2019](#)) - MOSFIRE (Keck) ([McLean et al., 2008](#)) - DEIMOS (Keck) ([Cowley et al., 1997](#)) - LRIS (Keck) ([Oke et al., 1995](#)) - GMOS (Gemini-N and Gemini-S) ([Davies et al., 1997](#)) - Flamingos2 (Gemini) ([Eikenberry et al., 2012](#)) - FourStar (Magellan) ([Persson et al., 2013](#)) - IMACS (Magellan) ([Bigelow & Dressler, 2003](#))

POTPyRI is meant to provide the community with a way to reduce imaging data from these instruments in a streamlined manner, using easily accessible code.

State of the field

Given the more complex suite of reductions often needed for spectroscopic data, many pipelines have been developed to reduce spectroscopic data from many instruments. However, at the start of development, no other automated pipeline was available to reduce images from a number of these instruments, with only guidelines and recipes often described. As the main research undertaken by the developers and collaborators often used data from these instruments,

39 a need to have a way to quickly and automatically reduce these data to provide information
40 on subsequent follow-up observations was born. Gemini has since set up DRAGONS (Labrie
41 et al., 2023) for that facility, including imaging; other projects offer instrument-specific or
42 spectroscopic-focused pipelines. We built POTPyRI rather than contributing only to DRAGONS
43 or a single-facility tool because our research required one pipeline for imaging from multiple
44 facilities (Keck, Gemini, MMT, Magellan) with consistent calibration and photometry, and
45 features such as optional relative flux calibration before stacking, Gaia-based astrometric
46 refinement, and both aperture and PSF photometry in one workflow. POTPyRI provides fine
47 astrometric fitting, photometric calibration, and aperture and PSF photometry on all detected
48 sources in the stack.

49 Software design

50 POTPyRI was developed in Python to take advantage of Python-based astronomy data
51 reduction and to move away from older “blackbox” and licensed programs. Key design trade-
52 offs: we use instrument-specific settings files rather than one global configuration, so each
53 telescope defines its own keywords and calibration options and new instruments can be added
54 without changing core code; we use external tools (SExtractor, Astrometry.net) for source
55 detection and initial WCS to improve robustness while the pipeline focuses on orchestration
56 and photometry; and we build on the Astropy ecosystem (ccdproc, photutils, astroquery)
57 for units, coordinates, combination, and catalog queries. The core reduction flow is shared
58 across instruments (calibration, masking, sky subtraction, astrometry, alignment, stacking,
59 photometry), with instrument subclasses supplying detector-specific parameters and file-naming.
60 POTPyRI is hosted on GitHub to encourage feedback and contributions.

61 Method

62 POTPyRI requires two parameters to run. The first is the name of the instrument in order to
63 load the correct settings file. This settings file allows new and user instruments to be added as
64 needed. The second is the full data path to the data to be reduced. The data are expected to
65 be in a particular format, including the file name and the number of extensions, as specified in
66 the setting file. For example, for Keck, the pipeline expects the data in the format in which it
67 is download from the Keck Observatory Archive. In this case, the user can specify that the
68 input data conform to the archive format using `-proc archive`. The details on data formats,
69 including scripts to download and sort data, can be found on GitHub.

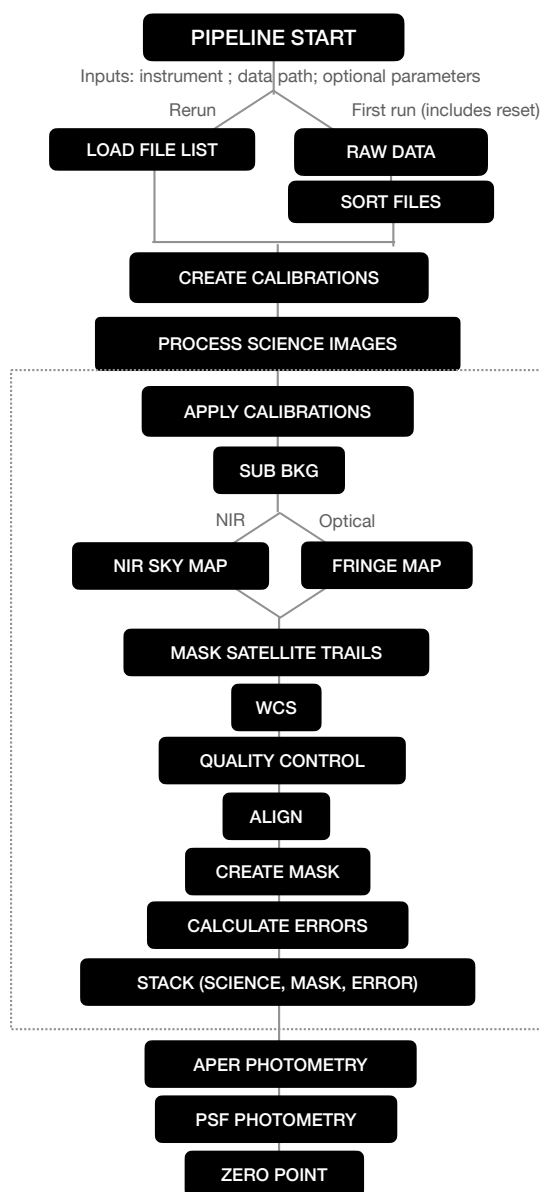


Figure 1: Flow diagram showing the basic steps taken by the pipeline. The dashed box shows all steps taken under the process science function, while all steps after the calibration creation is done per target.

Figure 1 shows the basic pipeline operations. The pipeline will first sort the files in the given path, and create a file list which contains information about each file such as the type, filter, configuration, and target. Using the file list and the settings file, the pipeline will then create calibration files (e.g. bias, dark, flat) and performs the necessary basic reductions to science files, grouped by target, including gain correction, bias subtraction, dark subtraction, flat correction, and trimming. During this step, a static mask, if available, is used to identify bad pixels. This mask is later extended to include image-specific issues such as satellite trails, cosmic rays (cleaned from images) and saturated stars. A 2D sky background is created for each image and subtracted. Astrometry.net is used on each image to provide a rough WCS solution, after which a fine astrometric solution is computed using Gaia-DR3 catalogs (Gaia Collaboration et al., 2021). Images are then aligned using the new WCS. After basic reductions, additional corrections where applicable are applied. For the NIR, the sky background can change during the night. Thus, for NIR observations, an additional NIR sky map is created

83 for each science image, based on observations for each target. In the optical, redder filters
84 such as *i* and *z*-bands, can also suffer from fringing, which introduces low-level structure in
85 the background. Thus, for filters that require this correction (based on each instrument and
86 thus set in the settings file for the instrument), the pipeline creates a fringe map based on all
87 science images for a target for that filter. An uncertainty map for each image is created to
88 include noise added by the reduction such as the background subtraction.

89 After solving for all the astrometry, the pipeline will remove any images with high astrometric
90 dispersion (i.e. images that have dispersion $> 5\sigma$ either in RA or DEC), except if the user
91 specified the `-keep_all_astro` option, before stacking images. Images are median stacked,
92 using the uncertainty masks and exposure times as weights, and a final stack mask is created
93 (see Figure [Figure 2](#) for examples of median stacks from MOSFIRE, MMIRS, LRIS (blue
94 channel), BINOSPEC (left side) and DEIMOS).

95 The pipeline will then perform both aperture and PSF photometry on sources in the stacked
96 image. For the aperture photometry, the pipeline will detect sources within the image and
97 determine statistics within a fiducial aperture radius, which is written as an extension to
98 the stacked image. For the PSF photometry, the PSF is determined using a list of bright
99 unsaturated stars, determined based on cuts on roundness, Full Width Half Maximum (FWHM),
100 and signal-to-noise. Next, the pipeline uses the PSF to calculate PSF photometry for all the
101 originally extracted sources. After the PSF photometry has been calculated, the pipeline will
102 then calculate a zero point based on aperture photometry by downloading a catalog of standard
103 stars. Currently supported catalogs are the Sloan Digital Sky Survey DR12 ([Eisenstein et al., 2011](#)),
104 Pan-STARRS ([Tonry et al., 2012](#)), SkyMapper ([Onken et al., 2024](#)), and 2MASS
105 ([Skrutskie et al., 2006](#)) covering most optical and infrared bands in the north and south. The
106 zero point is then transformed to the AB system and written to the header.

107 In addition, an approximate limiting magnitude is calculated for each stack using the FWHM
108 and the approximate standard deviation in the sky background per pixel. These values are
109 propagated into the image header as 3-, 5-, and 10- σ limiting magnitudes as *M3SIGMA*,
110 *M5SIGMA*, and *M10SIGMA*.

111 For more detailed instructions on running the pipeline and a description of pipeline outputs,
112 please see the documentation available on the GitHub. For user support or errors running the
113 pipeline, please open a Github issue.

114 While the pipeline runs, the pipeline will also create a detailed log of operations. Along with
115 the log, the pipeline will also write information such as the readnoise, effective gain, and
116 zero point to the stack. The pipeline also produces a number of other output files, including
117 source catalogs, star lists, error plots etc. For more details on these outputs, refer to the
118 documentation on GitHub.

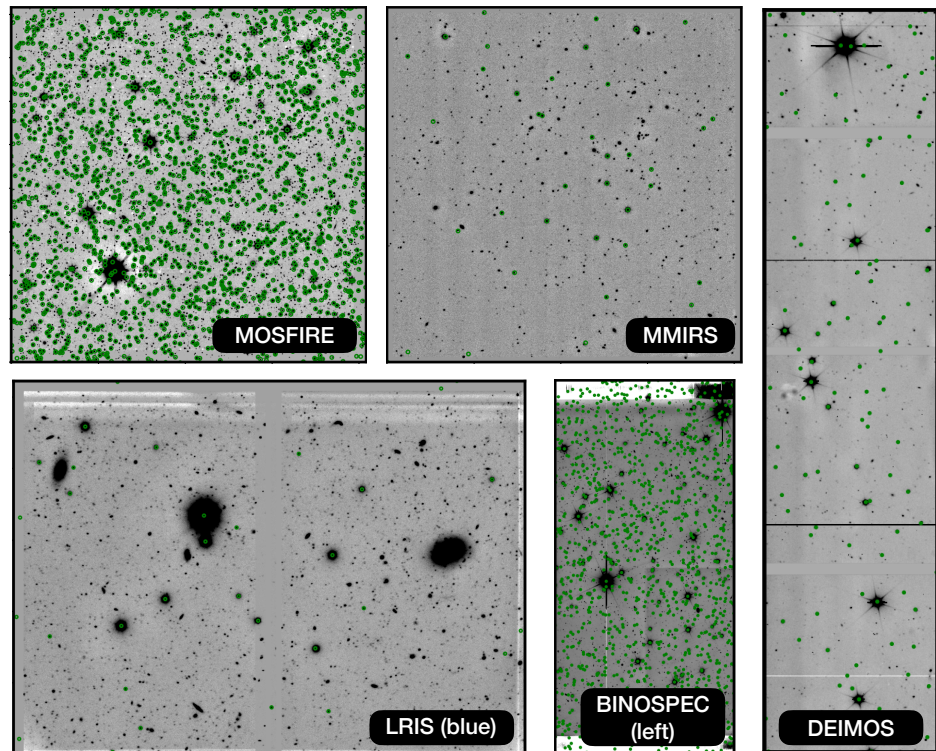


Figure 2: Examples of stacks produced by the pipeline from MOSFIRE, MMIRS, LRIS (blue side), BINOSPEC (left side) and DEIMOS data. The position of Gaia DR3 stars are shown by the green circles.

Research impact statement

POTPyRI has been used to reduce and stack imaging in published and in-preparation work on transient follow-up and host galaxy association, including Paterson et al. (2020), J. C. Rastinejad et al. (2021), Fong et al. (2021), Jillian C. Rastinejad et al. (2025), and Caleb et al. (2025). Applications include rapid reduction and identification of transients such as Gamma-Ray Burst (GRB) afterglows, and reduction and stacking of follow-up observations to identify host galaxies for association and study. The package is available on PyPI and is developed openly on GitHub with contributions from multiple developers. The pipeline continues to be improved while in use in ongoing projects by the developers and their collaborators.

AI usage disclosure

No AI tools were used in the building and preparation of this pipeline.

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References

- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., Günther, H. M., Lim, P. L., Crawford, S. M., Conseil, S., Shupe, D. L., Craig, M. W., Dencheva, N., Ginsburg, A., VanderPlas, J. T., Bradley, L. D., Pérez-Suárez, D., de Val-Borro, M., Aldcroft, T. L., Cruz, K. L., Robitaille, T. P., Tollerud, E. J., ... Astropy Contributors. (2018). The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package. *The Astronomical Journal*, 156(3), 123. <https://doi.org/10.3847/1538-3881/aabc4f>
- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., Greenfield, P., Droettboom, M., Bray, E., Aldcroft, T., Davis, M., Ginsburg, A., Price-Whelan, A. M., Kerzendorf, W. E., Conley, A., Crighton, N., Barbary, K., Muna, D., Ferguson, H., Grollier, F., Parikh, M. M., Nair, P. H., ... Streicher, O. (2013). Astropy: A community Python package for astronomy. *Astronomy & Astrophysics*, 558, A33. <https://doi.org/10.1051/0004-6361/201322068>
- Bertin, E., & Arnouts, S. (1996). SExtractor: Software for source extraction. *Astronomy and Astrophysics Supplement*, 117, 393–404. <https://doi.org/10.1051/aas:1996164>
- Bigelow, B. C., & Dressler, A. M. (2003). IMACS, the multiobject spectrograph and imager for Magellan: a status report. In M. Iye & A. F. M. Moorwood (Eds.), *Instrument design and performance for optical/infrared ground-based telescopes* (Vol. 4841, pp. 1727–1738). <https://doi.org/10.1117/12.461870>
- Bradley, L., Sipőcz, B., Robitaille, T., Tollerud, E., Vinícius, Z., Deil, C., Barbary, K., Wilson, T. J., Busko, I., Donath, A., Günther, H. M., Cara, M., Lim, P. L., Meßlinger, S., Conseil, S., Burnett, Z., Bostroem, A., Droettboom, M., Bray, E. M., ... Georgiev, I. Y. (2024). *Astropy/photutils: 2.0.2* (Version 2.0.2). Zenodo. <https://doi.org/10.5281/zenodo.13989456>
- Caleb, M., Nanayakkara, T., Stappers, B., Pastor-Marazuela, I., Khrykin, I. S., Glazebrook, K., Tejos, N., Prochaska, J. X., Rajwade, K., Mas-Ribas, L., Driessen, L. N., Fong, W., Gordon, A. C., Hoffmann, J., James, C. W., Jankowski, F., Kahinga, L., Kramer, M., Simha, S., ... Tian, J. (2025). A fast radio burst from the first 3 billion years of the Universe. *arXiv e-Prints*, arXiv:2508.01648. <https://doi.org/10.48550/arXiv.2508.01648>
- Cowley, D. J., Faber, S., Hilyard, D. F., James, E., & Osborne, J. (1997). DEIMOS: a wide-field faint-object spectrograph. In A. L. Ardeberg (Ed.), *Optical telescopes of today and tomorrow* (Vol. 2871, pp. 1107–1115). <https://doi.org/10.1117/12.268997>
- Davies, R. L., Allington-Smith, J. R., Bettess, P., Chadwick, E., Content, R., Dodsworth, G. N., Haynes, R., Lee, D., Lewis, I. J., Webster, J., Atad, E., Beard, S. M., Ellis, M., Hastings, P. R., Williams, P. R., Bond, T., Crampton, D., Davidge, T. J., Fletcher, M., ... Szeto, K. (1997). GMOS: the GEMINI Multiple Object Spectrographs. In A. L. Ardeberg (Ed.), *Optical telescopes of today and tomorrow* (Vol. 2871, pp. 1099–1106). <https://doi.org/10.1117/12.268996>
- Eikenberry, S., Bandyopadhyay, R., Bennett, J. G., Bessoff, A., Branch, M., Charcos, M., Corley, R., Dewitt, C., Eriksen, J.-D., Elston, R., Frommeyer, S., Gonzalez, A., Hanna, K., Herlevich, M., Hon, D., Julian, J., Julian, R., Lasso, N., Marin-Franch, A., ... Pazder, J. (2012). FLAMINGOS-2: the facility near-infrared wide-field imager and multi-object spectrograph for Gemini. In I. S. McLean, S. K. Ramsay, & H. Takami (Eds.), *Ground-based and airborne instrumentation for astronomy IV* (Vol. 8446, p. 84460I). <https://doi.org/10.1117/12.925679>
- Eisenstein, D. J., Weinberg, D. H., Agol, E., Aihara, H., Allende Prieto, C., Anderson, S. F., Arns, J. A., Aubourg, É., Bailey, S., Balbinot, E., Barkhouser, R., Beers, T. C., Berlind, A. A., Bickerton, S. J., Bizyaev, D., Blanton, M. R., Bochanski, J. J., Bolton, A. S., Bosman, C. T., ... Zhao, B. (2011). SDSS-III: Massive Spectroscopic Surveys of the Distant Universe, the Milky Way, and Extra-Solar Planetary Systems. *The Astronomical Journal*, 142(3), 72.

- 185 <https://doi.org/10.1088/0004-6256/142/3/72>
- 186 Fabricant, D., Fata, R., Epps, H., Gauron, T., Mueller, M., Zajac, J., Amato, S., Barberis, J.,
187 Bergner, H., Brennan, P., Brown, W., Chilingarian, I., Geary, J., Kradinov, V., McLeod,
188 B., Smith, M., & Woods, D. (2019). Binospec: A Wide-field Imaging Spectrograph for
189 the MMT. *Publications of the Astronomical Society of the Pacific*, 131(1001), 075004.
190 <https://doi.org/10.1088/1538-3873/ab1d78>
- 191 Fong, W., Laskar, T., Rastinejad, J., Escorial, A. R., Schroeder, G., Barnes, J., Kilpatrick, C.
192 D., Paterson, K., Berger, E., Metzger, B. D., Dong, Y., Nugent, A. E., Strausbaugh, R.,
193 Blanchard, P. K., Goyal, A., Cucchiara, A., Terreran, G., Alexander, K. D., Eftekhari, T., ...
194 Nicholl, M. (2021). The Broadband Counterpart of the Short GRB 200522A at $z = 0.5536$:
195 A Luminous Kilonova or a Collimated Outflow with a Reverse Shock? *The Astrophysical*
196 *Journal*, 906(2), 127. <https://doi.org/10.3847/1538-4357/abc74a>
- 197 Gaia Collaboration, Brown, A. G. A., Vallenari, A., Prusti, T., de Bruijne, J. H. J., Babusiaux,
198 C., Biermann, M., Creevey, O. L., Evans, D. W., Eyer, L., Hutton, A., Jansen, F., Jordi, C.,
199 Klioner, S. A., Lammers, U., Lindegren, L., Luri, X., Mignard, F., Panem, C., ... Zwitter,
200 T. (2021). Gaia Early Data Release 3. Summary of the contents and survey properties.
201 *Astronomy & Astrophysics*, 649, A1. <https://doi.org/10.1051/0004-6361/202039657>
- 202 Ginsburg, A., Sipőcz, B. M., Brasseur, C. E., Cowperthwaite, P. S., Craig, M. W., Deil, C.,
203 Guillochon, J., Guzman, G., Liedtke, S., Lian Lim, P., Lockhart, K. E., Mommert, M., Morris,
204 B. M., Norman, H., Parikh, M., Persson, M. V., Robitaille, T. P., Segovia, J.-C., Singer, L.
205 P., ... a subset of astropy Collaboration. (2019). astroquery: An Astronomical Web-querying
206 Package in Python. *The Astronomical Journal*, 157(3), 98. <https://doi.org/10.3847/1538-3881/aafc33>
- 207
- 208 Labrie, K., Simpson, C., Cardenes, R., Turner, J., Soraisam, M., Quint, B., Oberdorf, O.,
209 Placco, V. M., Berke, D., Smirnova, O., Conseil, S., Vacca, W. D., & Thomas-Osip, J.
210 (2023). DRAGONS-A Quick Overview. *Research Notes of the American Astronomical*
211 *Society*, 7(10), 214. <https://doi.org/10.3847/2515-5172/ad0044>
- 212 Lang, D., Hogg, D. W., Mierle, K., Blanton, M., & Roweis, S. (2010). Astrometry.net: Blind
213 Astrometric Calibration of Arbitrary Astronomical Images. *The Astronomical Journal*,
214 139(5), 1782–1800. <https://doi.org/10.1088/0004-6256/139/5/1782>
- 215 McLean, I. S., Steidel, C. C., Matthews, K., Epps, H., & Adkins, S. M. (2008). MOSFIRE:
216 a multi-object near-infrared spectrograph and imager for the Keck Observatory. In I. S.
217 McLean & M. M. Casali (Eds.), *Ground-based and airborne instrumentation for astronomy*
218 *II* (Vol. 7014, p. 70142Z). <https://doi.org/10.1117/12.788142>
- 219 McLeod, B., Fabricant, D., Nystrom, G., McCracken, K., Amato, S., Bergner, H., Brown, W.,
220 Burke, M., Chilingarian, I., Conroy, M., Curley, D., Furesz, G., Geary, J., Hertz, E., Holwell,
221 J., Matthews, A., Norton, T., Park, S., Roll, J., ... Martini, P. (2012). MMT and Magellan
222 Infrared Spectrograph. *Publications of the Astronomical Society of the Pacific*, 124(922),
223 1318. <https://doi.org/10.1086/669044>
- 224 Oke, J. B., Cohen, J. G., Carr, M., Cromer, J., Dingizian, A., Harris, F. H., Labrecque,
225 S., Lucinio, R., Schaal, W., Epps, H., & Miller, J. (1995). The Keck Low-Resolution
226 Imaging Spectrometer. *Publications of the Astronomical Society of the Pacific*, 107, 375.
227 <https://doi.org/10.1086/133562>
- 228 Onken, C. A., Wolf, C., Bessell, M. S., Chang, S.-W., Luvaul, L. C., Tonry, J. L., White, M.
229 C., & Da Costa, G. S. (2024). SkyMapper Southern Survey: Data release 4. *Publications*
230 *of the Astronomical Society of Australia*, 41, e061. <https://doi.org/10.1017/pasa.2024.53>
- 231 Paterson, K., Fong, W., Nugent, A., Escorial, A. R., Leja, J., Laskar, T., Chornock, R., Miller,
232 A. A., Scharwächter, J., Cenko, S. B., Perley, D., Tanvir, N. R., Levan, A., Cucchiara, A.,
233 Cobb, B. E., De, K., Berger, E., Terreran, G., Alexander, K. D., ... Cornish, D. (2020).

- 234 Discovery of the Optical Afterglow and Host Galaxy of Short GRB 181123B at $z =$
 235 1.754: Implications for Delay Time Distributions. *The Astrophysical Journal*, 898(2), L32.
 236 <https://doi.org/10.3847/2041-8213/aba4b0>
- 237 Persson, S. E., Murphy, D. C., Smee, S., Birk, C., Monson, A. J., Uomoto, A., Koch, E.,
 238 Shectman, S., Barkhouser, R., Orndorff, J., Hammond, R., Harding, A., Scharfstein, G.,
 239 Kelson, D., Marshall, J., & McCarthy, P. J. (2013). FourStar: The Near-Infrared Imager for
 240 the 6.5 m Baade Telescope at Las Campanas Observatory. *Publications of the Astronomical*
 241 *Society of the Pacific*, 125(928), 654. <https://doi.org/10.1086/671164>
- 242 Rastinejad, J. C., Fong, W., Kilpatrick, C. D., Paterson, K., Tanvir, N. R., Levan, A. J.,
 243 Metzger, B. D., Berger, E., Chornock, R., Cobb, B. E., Laskar, T., Milne, P., Nugent,
 244 A. E., & Smith, N. (2021). Probing Kilonova Ejecta Properties Using a Catalog of
 245 Short Gamma-Ray Burst Observations. *The Astrophysical Journal*, 916(2), 89. <https://doi.org/10.3847/1538-4357/ac04b4>
- 247 Rastinejad, Jillian C., Levan, A. J., Jonker, P. G., Kilpatrick, C. D., Fryer, C. L., Sarin, N.,
 248 Gompertz, B. P., Liu, C., Eyles-Ferris, R. A. J., Fong, W., Burns, E., Gillanders, J. H.,
 249 Mandel, I., Malesani, D. B., O'Brien, P. T., Tanvir, N. R., Ackley, K., Aryan, A., Bauer,
 250 F. E., ... Yang, S. (2025). EP 250108a/SN 2025kg: Observations of the Most Nearby
 251 Broad-line Type Ic Supernova Following an Einstein Probe Fast X-Ray Transient. *The*
 252 *Astrophysical Journal*, 988(1), L13. <https://doi.org/10.3847/2041-8213/ade7f9>
- 253 Skrutskie, M. F., Cutri, R. M., Stiening, R., Weinberg, M. D., Schneider, S., Carpenter, J.
 254 M., Beichman, C., Capps, R., Chester, T., Elias, J., Huchra, J., Liebert, J., Lonsdale, C.,
 255 Monet, D. G., Price, S., Seitzer, P., Jarrett, T., Kirkpatrick, J. D., Gizis, J. E., ... Wheelock,
 256 S. (2006). The Two Micron All Sky Survey (2MASS). *The Astronomical Journal*, 131(2),
 257 1163–1183. <https://doi.org/10.1086/498708>
- 258 Tonry, J. L., Stubbs, C. W., Lykke, K. R., Doherty, P., Shivvers, I. S., Burgett, W. S.,
 259 Chambers, K. C., Hodapp, K. W., Kaiser, N., Kudritzki, R.-P., Magnier, E. A., Morgan, J.
 260 S., Price, P. A., & Wainscoat, R. J. (2012). The Pan-STARRS1 Photometric System. *The*
 261 *Astrophysical Journal*, 750(2), 99. <https://doi.org/10.1088/0004-637X/750/2/99>